

ASHEVILLE 8 SSW, NC, USA

WMO: 746960

Lat: 35.4945N

Lon: 82.6142W

Elev: 2151

StdP: 13.59

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 53877

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.9	17.2	2.0	6.6	19.2	5.9	8.1	23.3	5.0	36.2	4.9	36.3	1.2	N/A	0.365

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.3	86.3	70.2	84.2	69.8	82.2	69.2	73.9	80.8	72.9	79.8	71.8	78.7	2.1	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.0	128.2	76.8	70.7	122.8	76.1	69.6	117.8	75.3	39.0	80.9	38.0	80.0	37.0	78.9	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
4.8	4.5	4.0	DB	5.0	89.8	4.8	2.9	1.5	91.9	-1.3	93.6	-4.0	95.2	-7.5	97.3	
			WB	4.1	75.7	5.0	1.3	0.5	76.7	-2.4	77.4	-5.3	78.2	-8.9	79.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.7	35.8	39.3	46.5	54.9	62.1	68.8	71.3	70.7	65.4	55.3	45.6	39.8
	DBStd	14.31	9.73	8.54	8.53	7.18	6.12	3.91	3.44	3.66	5.40	7.53	7.86	8.56
	HDD50	1516	447	312	172	33	4	0	0	0	1	35	177	335
	HDD65	4385	904	719	573	307	123	13	3	6	60	310	584	783
	CDD50	3235	8	13	63	181	379	565	661	643	462	200	43	17
	CDD65	631	0	0	0	5	33	128	200	183	71	10	1	0
	CDH74	4648	0	2	15	157	411	1002	1317	1161	510	69	4	0
	CDH80	961	0	0	0	17	52	220	315	260	90	7	0	0
Wind	WSAvg	1.2	1.6	1.7	1.8	1.7	1.2	0.9	0.7	0.7	0.7	1.0	1.3	1.4
Precipitation	PrecAvg	59.0	4.9	4.6	5.4	4.7	4.3	5.6	5.5	5.6	5.2	3.8	4.2	5.3
	PrecMax	84.6	11.7	9.3	10.1	8.7	10.0	10.6	14.2	10.1	19.3	8.8	8.6	10.2
	PrecMin	39.7	1.3	1.9	2.0	2.1	1.5	1.6	1.9	1.0	0.9	0.3	1.5	2.3
	PrecStd	10.1	2.5	1.8	1.9	1.6	2.1	2.2	2.7	2.2	3.5	2.2	1.8	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.5	71.4	76.3	82.5	84.5	88.3	88.5	88.6	86.1	80.2	73.5	68.1
		MCWB	55.4	57.7	57.4	60.8	65.9	69.2	71.7	70.8	68.1	67.2	59.9	57.5
	2%	DB	61.2	64.5	71.7	78.4	81.4	85.1	85.9	85.6	82.6	75.5	68.3	63.0
		MCWB	53.1	53.3	56.4	60.1	65.3	69.4	71.2	71.2	68.0	62.5	56.0	56.0
	5%	DB	56.9	60.0	67.5	75.1	78.7	82.4	83.7	83.0	80.0	72.0	64.6	58.9
		MCWB	50.0	50.0	54.2	58.5	64.5	68.8	71.1	70.7	67.6	61.9	55.7	53.0
	10%	DB	52.0	55.2	63.1	71.5	75.9	79.9	81.4	80.4	76.9	68.8	61.2	54.7
		MCWB	47.5	46.8	51.9	57.2	63.7	68.1	70.8	70.1	66.7	60.6	53.3	49.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.7	60.4	60.7	65.3	70.4	73.7	76.0	76.0	72.4	70.3	64.5	61.1
		MCDB	64.2	67.9	70.5	75.6	78.0	80.6	82.2	81.9	79.4	76.3	67.5	65.0
	2%	WB	55.1	56.8	58.8	63.1	68.5	72.4	73.9	74.1	70.9	67.4	60.7	57.7
		MCDB	59.6	62.4	67.6	73.2	76.2	79.6	80.9	80.8	78.1	72.4	65.1	61.6
	5%	WB	51.4	52.2	56.6	61.1	67.2	71.2	72.9	72.9	69.4	65.0	58.3	54.0
		MCDB	54.4	57.4	64.4	70.2	74.8	78.6	80.1	79.6	75.8	69.2	62.7	57.6
	10%	WB	47.8	48.2	54.0	59.3	65.6	69.8	71.9	71.6	68.2	62.5	54.9	50.3
		MCDB	51.2	53.6	60.5	67.9	73.0	77.2	78.8	78.1	74.2	66.7	59.4	53.8
Mean Daily Temperature Range	5% DB	MDBR	21.9	22.3	23.8	25.9	23.7	22.4	20.3	20.1	20.9	23.4	25.0	20.9
		MCDBR	29.6	31.2	31.8	32.9	28.5	26.1	23.5	24.0	25.8	27.3	30.3	25.8
		MCWBR	20.1	20.0	17.7	16.5	14.3	12.6	11.2	11.6	12.6	15.3	18.2	17.6
	5% WB	MCDBR	22.3	24.9	25.7	26.0	22.2	22.0	20.5	20.0	19.5	18.3	20.1	20.9
		MCWBR	17.7	18.3	15.9	14.3	12.4	11.7	10.5	10.5	10.5	11.7	14.6	16.3
Clear-Sky Solar Irradiance	taub	0.294	0.303	0.332	0.372	0.426	0.451	0.492	0.490	0.385	0.351	0.313	0.291	
	taud	2.545	2.513	2.457	2.346	2.210	2.172	2.087	2.072	2.352	2.442	2.516	2.581	
	Ebn at Noon	291	300	298	289	273	266	254	252	276	278	280	285	
	Edh at Noon	25	29	33	39	46	47	51	51	36	31	25	23	
All-Sky Solar Radiation	RadAvg	816	1014	1345	1666	1831	1925	1793	1681	1458	1215	914	704	
	RadStd	51	108	86	127	135	134	126	79	130	125	80	63	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+250	N/A
(47)	Regional (40 neighbors)	+0.97	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+277	+180

Nomenclature: See separate page